

Effectiveness of MR to enhance drivers' perception of turnability

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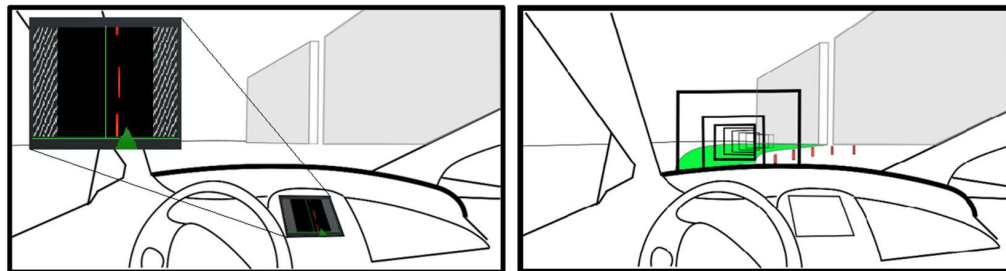
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Figure 1 : (1) Traditional configural display and (2) Conformal (i.e., MR) display designed to enhance the perception of turnability

ABSTRACT

The automotive industry faces the challenge of determining whether Mixed Reality displays (MR), which are expensive, outperform traditional head-down displays in depicting Advanced Driver Assistance Systems (ADAS). We designed a novel ecological interface for an ADAS to enhance drivers' perception of Turnability - an affordance specifying the possibility of crossing a narrow orthogonal alleyway following a forward constant-curvature turn. We augmented the perception of turnability with configural (head-down) and conformal (MR) displays. Overall, both displays improved drivers' decision-making. However, the configural display is not only superior but also rated as more usable than the conformal display by the drivers.

Index terms: Turn, Affordance, Ecological Interface Design, Conformal, Configural.

1 INTRODUCTION

The tight turns required to weave through congested spaces (urban core, parking lots, garages) put drivers' perception of their car's maneuverability to the test. The car's Ackermann design limits the maximum curvature trajectory it can follow (k_{max}). This maneuverability limit divides the curvature required to turn toward a narrow orthogonal alleyway (k_{req}) into possible and impossible ranges. In other words, the dimensionless ratio π between k_{req} and k_{max} expresses the turning-ability -or "turnability"- affordance, which specifies the possibility of turning into an alleyway. Modeling turnability as an affordance is an asset to exploit the Ecological Interface Design (EID) approach (Vicente & Rasmussen, 1990) in the design of new Advanced Driver Assistance Systems (ADAS) [1]. However, it remains an open question whether drivers would benefit more from an enhancement of turnability through a head-down display with a configural display [2] or through a Mixed Reality (MR) interface with a conformal display [3].

Similarly to other action-scaled affordances, turnability can be visualized as a landscape [4] by launching tentacles as ego-centered trajectories emanating from the vehicle, whose curvature changes with the steering wheel. Depicting this landscape (Figure 2A) in an ADAS through a configural (Figure 2B) or a conformal (Figure 2C) display would allow drivers to visualize whether they can, when and how initiating a turn to cross the alleyway. The landscape is

colored by a custom-made "crossing score" - indicator which quantifies the distribution of lateral margins when crossing and helps drivers in selecting among multiple crossing possibilities.

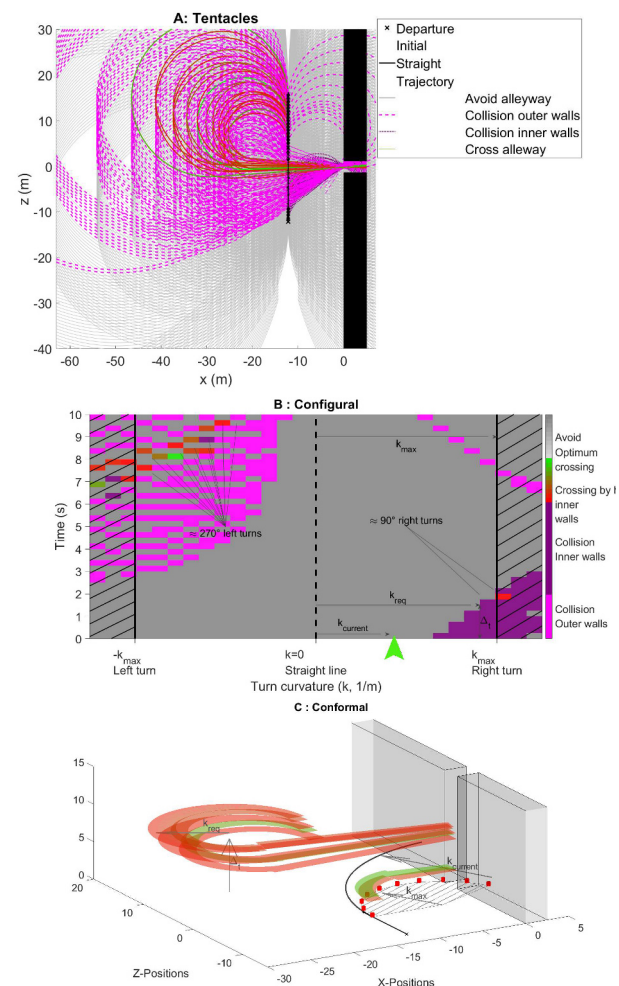


Figure 2: (A) Tentacles shape the turnability landscape colored by the "crossing score". Turnability can be visualized in a (B) traditional configural and (C) conformal (i.e., MR) displays.

2 METHODS

25 drivers were immersed in an urban environment in a fixed-base driving simulator. The first calibration phase allowed them to tackle the vehicle's dimensions and handling limits. During the experiment, drivers were asked to decide whether they could turn at a forward constant speed of 10 km/h to cross a narrow orthogonal alleyway surrounded by two buildings. Drivers responded before the turn (i.e., when the vehicle remained stationary for 7 seconds), and during the turn (i.e., when the vehicle was moving forward). None of the drivers reported experiencing motion sickness during the experiment, as assessed using the Fast Motion Sickness Scale.

The turnability π was manipulated by varying the initial position of the vehicle relative to the alleyway (i.e., varying k_{req}) while keeping k_{max} of the virtual car constant. We manipulated the availability of the configural (head-down) and conformal (MR) display with respect to a no-augmentation condition.

PSE (i.e., the critical π value from which drivers perceived the turn as impossible) were individually and separately extracted from logistic fits of turn frequency as a function of π , before, and during the turn. The usability of visual displays was assessed with the French version of the System Usability Scale, F-SUS [5].

3 RESULTS

Red plots in Figure 3 indicate that before the turn, both configural and conformal displays enhanced decision-making compared to the no-augmentation. A Fligner-Killeen test on PSE values ($\chi^2 = 35.44$; $df = 2$; $p < 0.01$) and subsequent pairwise comparisons revealed a significantly reduced variance with Configural ($\chi^2 = 19.97$; $df = 1$; $p_{adj} < 0.01$) and Conformal displays ($\chi^2 = 23.35$; $df = 1$; $p_{adj} < 0.01$), versus no-augmentation.

Blue plots show that during the turn, only the configural display improved decision-making. A Friedman test on individual PSE values ($n = 25$; $\chi^2 = 14$; $df = 2$; $p < 0.05$; $W = 0.28$) with Nemenyi post-hoc tests revealed significantly higher PSE values for the configural display compared to the no-augmentation ($p < 0.05$).

Green plots show that before the turn, the configural display was rated as more usable than the conformal display. A two-tailed Wilcoxon paired rank test ($n = 25$; $Z = -2.44$; $p < 0.05$; $r = 0.49$) showed significantly better F-SUS scores for the configural display ("Good" < 82.5) versus the conformal display ("OK" < 65.0).

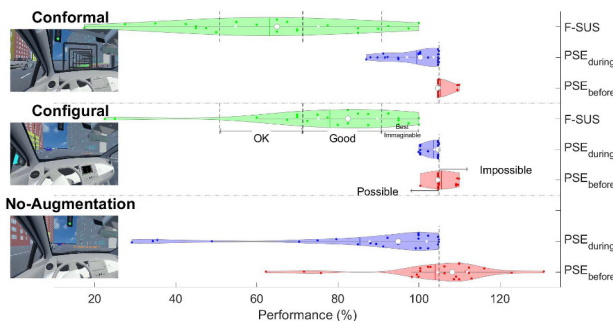


Figure 3: Benefits on decision-making were assessed with PSE measured before (red) and during the turn (blue). Usability was evaluated through F-SUS scores (in green).

4 DISCUSSION

Through the prism of the EID approach, we compared the benefits offered to drivers by configural and conformal visual enhancement of the affordance of turnability. While the configural display improved decision-making both before and during the turn, the conformal display only did it before the turn. This suggests that

both displays effectively convey to the driver's eye action-scaled affordance [6, 7]. Moreover, the configural display was rated as more usable than the conformal display. Superiority of configural display is consistent with historical works on EID [8], as well as with more recent results obtained with configural augmentations designed for automotive [1] and aeronautics domain [4, 9]. However, these benefits could vary depending on the use-case [10].

5 CONCLUSION

Mixed Reality display doesn't outperform traditional head-down display for enhancing the perception of turnability. This conclusion should be reassessed by extending this work to complex turns, drift turns, and other domains of vehicle turn control.

SUPPLEMENTAL MATERIALS

Two movies show successful turn performed with the Configural and Conformal displays.

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